

ShopSense Analytics Platform

Multi-Vendor E-Commerce Decision Intelligence & AI Analytics System

1. EXECUTIVE SUMMARY

ShopSense is an enterprise-grade Multi-Vendor E-Commerce Analytics and Decision Intelligence Platform. While conventional e-commerce applications focus strictly on transactional cart checkouts, ShopSense addresses the operational challenges faced by multi-merchant marketplaces. It converts raw transactional database records into dynamic business intelligence, automated inventory run-rate forecasts, customer lifetime value (LTV) segmentation, heuristic recommendation leaderboards, and Retrieval-Augmented Generation (RAG) artificial intelligence diagnostics via Google Gemini API.

2. PROBLEM STATEMENT

Modern multi-vendor e-commerce marketplaces (such as Amazon or Flipkart) host thousands of independent merchant storefronts. However, smaller merchant vendors often suffer from severe operational handicaps:

- **Unpredictable Stockouts:** Merchants lack automated forecasting tools, leading to depleted physical stock for high-demand catalog items.
- **Opaque Customer Retention:** Vendors cannot identify high-value repeat buyers versus single-purchase customers, hindering targeted retention.
- **Manual Reporting Overhead:** Processing weekly sales numbers requires tedious manual spreadsheet calculations.
- **Catalog Copywriting Barriers:** Drafting SEO-optimized product titles, descriptions, and setting competitive prices remains time-consuming.

3. PROPOSED SYSTEM ARCHITECTURE

ShopSense enforces a strict three-tier decoupled architectural pattern separating API controller routers from core domain service modules and relational database entities:

System Tier Decomposition

- **Controller Layer (app/routers/):** Lightweight HTTP handlers using FastAPI for URL path routing, Pydantic schema validation, and HTTP status handling.
- **Domain Service Layer (app/services/):** Encapsulated business logic for inventory forecasting, customer segmentation, recommendation heuristics, streaming CSV generation, and AI prompt engineering.
- **Data Persistence Layer (app/models/):** Relational SQLAlchemy ORM schemas mapping Admin, Vendor, Customer, Product, and Transaction entities.
- **Reactive Frontend Layer (React + Tailwind CSS):** Single-page user interface featuring custom SVG trendlines, benchmarking cards, and real-time WebSocket streams.

4. DETAILED MODULE SPECIFICATIONS

MODULE 1

Dual-Token Authentication & Role-Based Access Control (RBAC)

Secure dual-token authentication utilizing short-lived JWT Access Tokens (15-minute expiration) and long-lived Refresh Tokens (7-day expiration) stored in HTTP-Only, SameSite cookies. Features an Admin Vendor Approval workflow and vendor-level data isolation.

MODULE 2

Warehouse Inventory Intelligence & Demand Forecasting

Monitors warehouse stock in real-time against safety thresholds (5, 10, 15 units). Calculates daily sales velocity (units/day) to output projected 30-day stockout run-rates, reorder urgency levels (Immediate, Upcoming, Optimal), and atomic restock APIs.

MODULE 3

SQL-Based Customer RFM Behavioral Clustering

Evaluates customer purchasing history across Recency, Frequency, and Monetary (RFM) dimensions. Automatically classifies buyers into VIP (expenditure ≥ 500), Regular (expenditure 150 to 500), New (expenditure < 150), and Inactive cohorts.

MODULE 4

Multi-Tier Heuristic Recommendation Engine

Implements three algorithmic recommendation modules: Category Sales Bestsellers (units sold and stock weighting), Frequently Bought Together Cross-Sells (co-purchase affinity scoring), and Personalized Customer Recommendation Feeds.

MODULE 5

Executive Business Intelligence & Streaming CSV Exports

Processes sales trends for SVG trendline charts, computes category revenue share, and calculates vendor benchmarking metrics (AOV, Revenue/Item, Stockout Ratio, Sales Velocity) against marketplace averages. Streams memory-efficient CSV reports for Sales, Inventory, and Customer profiles.

MODULE 6

Generative AI & RAG Integration (Google Gemini API)

Integrates Google Gemini API (gemini-3.5-flash) for RAG AI Shopping Assistants (grounded catalog product context search) and AI Executive Store Advisors (automated merchant store diagnostic audits and 3-part strategic action plans). Features intelligent local RAG fallback.

MODULE 7

Real-Time WebSocket Event Hub & AI Merchant Tools

Provides bi-directional WebSocket connection (/ws/events) broadcasting live marketplace order creation events. Includes an AI Product Copywriter (drafting SEO titles and descriptions from notes) and an AI Smart Price Optimizer (evaluating price elasticity).

5. DATA GOVERNANCE & UPTIME ASSURANCE

ShopSense enforces strict multi-tenant data isolation. When a vendor requests store analytics, the backend automatically extracts the authenticated vendor ID and scopes all database queries to `Transaction.vendor_id == vendor_id`. Furthermore, if external AI services are unreachable, the system automatically falls back to local database keyword indexing and rule-based diagnostic heuristics, ensuring 100% platform availability.